

COURSE DESCRIPTION

Ignition is a web-deployed SCADA platform built on OPC UA, used across manufacturing, water/wastewater, and energy sectors. This course covers the gateway architecture, tag system design, Perspective HMI development, Jython scripting, alarm notification pipelines, Tag Historian, database integration, and security configuration — building a complete SCADA application from device connection to management report.

PREREQUISITES

Basic SQL knowledge is helpful. No prior SCADA or HMI experience required.

COURSE TOPICS

1. **Gateway Architecture:** Installation, module system, web management interface, unlimited licensing model, and backup/restore procedures.
2. **Tag System:** OPC, expression, and memory tag types; UDT templates for repeated equipment; tag database as the system foundation.
3. **OPC-UA Integration:** Ignition as OPC-UA client and server, device driver configuration, and PLC-to-live-tags workflow.
4. **Perspective Module:** Browser-based HMI architecture, component library, View/Page/Session structure, and tag binding without custom code.
5. **Scripting:** Jython on tag events, gateway events, and schedules; system library access to tags, databases, and alarm pipelines.
6. **Alarm Notification:** Alarm modes, notification pipelines, escalation rules, roster configuration, and email/SMS delivery infrastructure.
7. **Tag Historian:** Storage providers, partitioning strategy, aggregation modes, and Store and Forward for gap-free historical data.
8. **Database Integration:** Named queries, Transaction Groups, and the path from field sensor value to SQL table row.
9. **Security and Access Control:** Identity providers, role mapping, project permissions, and SSL/TLS configuration across single-site and multi-site deployments.
10. **Field Lab:** End-to-end SCADA build — OPC-UA device connection, tag configuration, Perspective dashboard, alarm pipeline, historian, management report.

LEARNING OUTCOMES

- Connect an OPC-UA device and expose live tags in a functioning Ignition project.
- Build a Perspective dashboard with live trend displays, alarm status, and navigation.
- Configure an alarm notification pipeline with escalation rules and SMS/email delivery.
- Write Jython scripts for tag event handling, scheduled tasks, and database queries.
- Query historian data and generate a management report from stored process values.

REQUIRED SOFTWARE

- **Ignition trial** (free from [inductiveautomation.com](https://www.inductiveautomation.com), all platforms) — fully featured SCADA platform that resets every two hours; sufficient for all lab exercises.